# Flower Classification

https://github.com/aiyshwary/flower-classification-project

Python TensorFlow Keras OpenCV scikit-learn Matplotlib Seaborn

## OVERVIEW

A CNN-based image classification project that identifies five flower types (Daisy, Dandelion, Rose, Sunflower, Tulip) using the Kaggle Flowers Recognition dataset and a custom TensorFlow/Keras model pipeline.

## IMPACT

Built an end-to-end notebook workflow that prepares the dataset, trains a custom CNN, and evaluates multi-class accuracy to classify flower species from images.

## TECHNICAL WALKTHROUGH

### Dataset & Setup

Used the Kaggle Flowers Recognition dataset with five classes: Daisy, Dandelion, Rose, Sunflower, and Tulip. The dataset is organized into class folders and loaded into the notebook for training and validation.

- Install dependencies: TensorFlow, OpenCV, scikit-learn, matplotlib, seaborn.
- Place the dataset under flowers/<class_name>/ to match the expected directory structure.

### Preprocessing Pipeline

Images are resized to a consistent shape, normalized to [0,1], and split into train/validation sets to ensure fair evaluation across classes.

- Resize and normalize images to stabilize training.
- Create stratified train/validation splits for balanced class coverage.

### Model Architecture

Implemented a custom CNN in Keras with stacked convolution + pooling blocks followed by dense layers for multi-class classification.

- Convolutional blocks learn spatial features of petals and textures.
- Dense layers map extracted features to 5-class softmax outputs.

### Training & Evaluation

Tracked training and validation curves to monitor convergence and prevent overfitting.

- Visualized accuracy/loss trends to validate model stability.
- Evaluated class-wise performance using predictions on validation data.

### Reproducible Notebook

The full workflow is implemented in the notebook for end-to-end reproducibility and easy experimentation.

## KEY DETAILS

- Organized the Kaggle Flowers Recognition dataset into class-specific folders for five flower categories.
- Preprocessed images by resizing and normalizing pixel values, then created train/validation splits for model evaluation.
- Implemented a custom CNN architecture in TensorFlow/Keras for 5-class image classification.
- Tracked training/validation accuracy and loss across epochs and visualized results with Matplotlib/Seaborn.
- Packaged the workflow into a reproducible Jupyter Notebook for easy experimentation and iteration.